

Tutorial for Cyber-Physical Systems - Discrete Models

Exercise Sheet 13

⚠ The sum of achievable points in all 13 exercise sheets is 203 (+ 36 bonus points). To pass the tutorial (the "Studienleistung" of the module), it is sufficient to have presented 1 task in the tutorial sessions and to achieve 101 points.

On Thursday, January 29th, we will upload an additional (short) bonus sheet about next week's lecture. Although there will be no tutorial session for the bonus sheet, your tutors will provide feedback as usual. ⚠

Exercise 1: LTL Properties

12 Points

The goal of this task is to get familiar with LTL-semantics and to learn how to interpret LTL formulas over transition systems.

Given the following LTL properties over $AP = \{a, b, c\}$:

$$\varphi_1 = a \wedge \bigcirc b$$

$$\varphi_3 = \neg(a \cup \Box b)$$

$$\varphi_5 = \Diamond \Box a$$

$$\varphi_2 = a \cup b$$

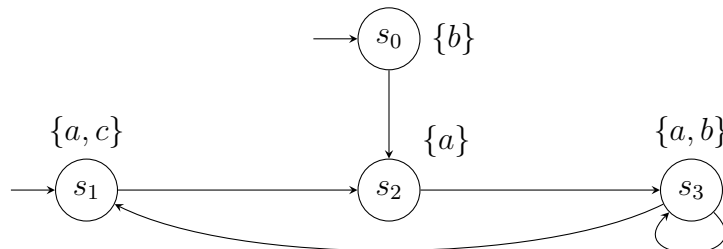
$$\varphi_4 = (\Diamond c) \cup \Box a$$

$$\varphi_6 = \Box \Diamond c$$

For each of the LTL properties φ_i complete the following tasks:

- (a) Give a trace $\tau \in (2^{AP})^\omega$ that satisfies φ_i .
- (b) Give a trace $\tau \in (2^{AP})^\omega$ that violates φ_i .
- (c) Determine if the transition system below satisfies φ_i . Explain why or why not.
- (d) Formalize $Words(\varphi_i)$ (i.e. the set of all traces satisfying φ_i) using set comprehension.

For example for $\varphi = \Diamond a$ we can formalize $Words(\varphi) = \{A_0 A_1 \dots \in (2^{AP})^\omega \mid \exists i \in \mathbb{N}. a \in A_i\}$.



Exercise 2: Stating properties in LTL

4 Points + 1 Bonus Points

The goal of this task is to learn to formally specify key properties in LTL.

Suppose we have two users, *Betsy* and *Peter*, and a single printer device. Both users perform several tasks, and every now and then they want to print their results on the printer. Since there is only a single printer, only one user can print a job at a time. Suppose we have the following atomic propositions for *Peter* at our disposal:

Peter.request indicates that *Peter* requests usage of the printer.

Peter.use indicates that *Peter* uses the printer.

Peter.release indicates that *Peter* releases the printer.

For *Betsy*, analogous predicates are defined. Specify in LTL the following properties:

- (a) Mutual exclusion, i.e., only one user at a time can use the printer.
- (b) Finite time of usage, i.e., a user can print only for a finite amount of time.
- (c) Absence of individual starvation, i.e., if a user wants to print something, the user is eventually able to do so.
- (d) Absence of blocking, i.e., if a user requests access to the printer, the user does not request forever.
- (e) **Bonus:** Alternating access, i.e., users must strictly alternate in printing.

Exercise 3: From Set Notation to LTL

4 Points + 1 Bonus Points

The goal of this exercise is to practice converting from set notation to LTL formulas.

Let $AP = \{a, b\}$. For each of the following properties P_i , give an equivalent LTL formula φ_i (if possible), i.e. give an LTL formula φ_i s.t. $P_i = \text{Words}(\varphi_i)$. For exactly one of the properties below, it is not possible to give an equivalent LTL formula. Identify this property. **Bonus:** Explain why it is not possible.

- (a) $P_1 = \{ A_0 A_1 \dots \in (2^{AP})^\omega \mid \exists i \in \mathbb{N}. a \in A_i \wedge b \in A_{i+1} \}$
- (b) $P_2 = \{ A_0 A_1 \dots \in (2^{AP})^\omega \mid \forall i \in \mathbb{N}. a \in A_{2i} \}$
- (c) $P_3 = \{ A_0 A_1 \dots \in (2^{AP})^\omega \mid \forall i \in \mathbb{N}. A_i = A_{i+2} \}$
- (d) $P_4 = \{ A_0 A_1 \dots \in (2^{AP})^\omega \mid \forall i \in \mathbb{N}. a \in A_i \rightarrow \exists j \in \mathbb{N}. j > i \wedge b \in A_j \}$